Reliability-Weighted Multimodal AI for Early Autism Risk Screening

Ten hybrid deep-learning architectures, 10-fold cross-validation:
test-fold results, statistical and shortcut tests, and overall final result

BITE308L Artificial Intelligence (VIT)
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Screening-support research only. No result in this document is a diagnosis or an estimate of clinical accuracy. All numbers are out-of-fold results on held-out test folds and were re-verified independently before this document was produced.

Summary

We trained and compared ten hybrid deep-learning architectures (a pretrained or custom-CNN spatial encoder, a recurrent or Transformer temporal encoder and an attention head) for autism risk screening on two public datasets: eye-tracking recordings of 57 children (D01) and 2,940 web-collected facial images (D02). All models used one fixed training protocol and grouped 10-fold cross-validation; tabular baselines were run on three questionnaire datasets. On D01, child-level pooled ROC-AUC ranged from 0.574 to 0.832, but a logistic regression on three recording-quality features alone reached 0.823. No model was significantly better than this baseline, and only one model (M07) had a matched-subset AUC whose 95% bootstrap CI excluded 0.5 (0.724 [0.551, 0.883], 14 matched pairs); the other models' matched AUCs had CIs including 0.5. On D02, models reached mean ROC-AUCs of 0.867-0.931, yet simple non-facial image statistics alone reached 0.796, so dataset artefacts separate the classes. Removing near-duplicate test images changed AUCs by at most 0.002. Pretrained encoders scored higher on average than custom CNNs on both datasets. The results do not show that the models learned autism-related gaze or facial patterns and are not clinical accuracy estimates; they motivate fusion that weights each modality by its estimated reliability.

1. What was trained and how it was tested

Data. D01 eye-tracking: 57 children (27 ASD / 30 TD), 1,241 clips of 8 one-second gaze images. D02 facial images: 2,940 web-collected photos (1,470 / 1,470), 2,929 near-duplicate groups. Three questionnaire tables (D04, D05, D06) for non-deep baselines.

Models. Ten hybrids M01-M10: a spatial encoder (seven ImageNet-pretrained timm backbones or a custom CNN trained from scratch for M02, M05, M09) -> linear projection -> temporal encoder (LSTM / BiLSTM / GRU / Transformer) -> attention pooling -> P(ASD). D01 sequences run over time, D02 sequences over spatial tokens.

Model	Architecture	Model	Architecture
M01	EfficientNetB3 + BiLSTM + Attn	M06	DenseNet121 + LSTM + Attn
M02	CNN + BiLSTM + Attn	M07	ViT + LSTM + Attn
M03	ResNet50 + LSTM + Attn	M08	EfficientNet + Transformer + Attn
M04	ConvNeXt + BiLSTM + Attn	M09	CNN + GRU + Attn
M05	CNN + Transformer + MLP	M10	ResNet50 + BiLSTM + Transformer

Protocol (fixed, never tuned on test data). AdamW, cosine schedule with warm-up, at most 40 epochs, early stopping on validation ROC-AUC (patience 8), mixed precision, batch 64, seed 42 + fold. Grouped, stratified 10-fold cross-validation (D01 by child, D02 by near-duplicate group); in every fold a validation part of the training groups chooses the stopping epoch and the threshold, and the test fold is predicted exactly once.

Scale. 200 deep-learning runs (10 models x 2 datasets x 10 folds) on one NVIDIA RTX A2000 12 GB: 43.8 min GPU time for D01 and 208.5 min for D02 (4.20 h in total), plus 200 tabular runs (5 models x 4 settings x 10 folds) on CPU.

F1. Hybrid model pipeline (one model id = one spatial encoder + one temporal encoder + one head)

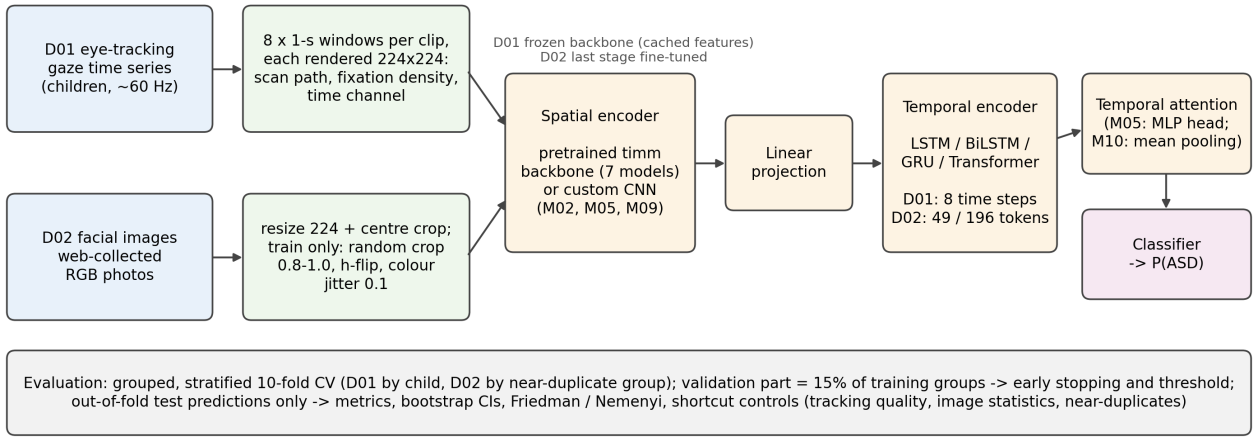


Figure 1. Hybrid model pipeline and evaluation design.

Reading the metrics. ROC-AUC mean ± SD is over the 10 test folds; pooled AUC is one AUC over all out-of-fold test predictions with a 95% bootstrap confidence interval (2,000 resamples of children / near-duplicate groups). Balanced accuracy, sensitivity and specificity use threshold 0.5. ECE = expected calibration error (lower is better).

2. Results on the held-out test folds

2.1 D01 eye-tracking, child level (one score per child = mean clip probability)

Model	ROC-AUC mean ± SD	Pooled AUC [95% CI]	Bal. acc.	Sensitivity	Specificity	Brier	ECE
M04	0.856 ± 0.137	0.762 [0.628, 0.883]	0.675 ± 0.121	0.617 ± 0.236	0.733 ± 0.378	0.208	0.164
M08	0.828 ± 0.210	0.786 [0.658, 0.895]	0.717 ± 0.143	0.600 ± 0.251	0.833 ± 0.283	0.190	0.121
M07	0.800 ± 0.219	0.832 [0.722, 0.927]	0.758 ± 0.186	0.617 ± 0.334	0.900 ± 0.161	0.188	0.158
M06	0.783 ± 0.207	0.784 [0.656, 0.895]	0.675 ± 0.133	0.617 ± 0.236	0.733 ± 0.263	0.197	0.116
M05	0.767 ± 0.187	0.605 [0.443, 0.756]	0.575 ± 0.127	0.550 ± 0.416	0.600 ± 0.466	0.266	0.217
M10	0.761 ± 0.193	0.673 [0.521, 0.809]	0.617 ± 0.177	0.567 ± 0.446	0.667 ± 0.471	0.231	0.119
M01	0.761 ± 0.219	0.714 [0.569, 0.840]	0.692 ± 0.201	0.550 ± 0.352	0.833 ± 0.283	0.219	0.195
M03	0.744 ± 0.195	0.735 [0.604, 0.852]	0.675 ± 0.114	0.517 ± 0.277	0.833 ± 0.324	0.211	0.145
M09	0.706 ± 0.203	0.574 [0.427, 0.725]	0.500 ± 0.000	0.767 ± 0.417	0.233 ± 0.417	0.357	0.366
M02	0.678 ± 0.283	0.578 [0.425, 0.731]	0.483 ± 0.053	0.833 ± 0.324	0.133 ± 0.322	0.285	0.244

2.2 D01 eye-tracking, clip level

Model	ROC-AUC mean ± SD	Pooled AUC [95% CI]	Bal. acc.	Sensitivity	Specificity	Brier	ECE
M08	0.686 ± 0.128	0.689 [0.594, 0.781]	0.605 ± 0.092	0.503 ± 0.249	0.707 ± 0.177	0.218	0.121
M04	0.683 ± 0.081	0.657 [0.560, 0.747]	0.614 ± 0.057	0.552 ± 0.193	0.676 ± 0.219	0.227	0.113
M07	0.678 ± 0.099	0.675 [0.610, 0.745]	0.630 ± 0.092	0.522 ± 0.183	0.738 ± 0.096	0.221	0.096
M05	0.657 ± 0.115	0.610 [0.456, 0.748]	0.543 ± 0.059	0.481 ± 0.385	0.605 ± 0.366	0.276	0.238
M01	0.645 ± 0.108	0.634 [0.544, 0.717]	0.588 ± 0.083	0.484 ± 0.130	0.692 ± 0.206	0.247	0.146
M06	0.639 ± 0.108	0.652 [0.576, 0.728]	0.596 ± 0.083	0.481 ± 0.157	0.710 ± 0.143	0.234	0.125
M10	0.638 ± 0.134	0.628 [0.509, 0.746]	0.573 ± 0.096	0.557 ± 0.319	0.589 ± 0.343	0.262	0.195
M03	0.635 ± 0.107	0.625 [0.542, 0.707]	0.579 ± 0.090	0.416 ± 0.131	0.741 ± 0.203	0.245	0.115
M09	0.605 ± 0.087	0.541 [0.398, 0.693]	0.518 ± 0.026	0.726 ± 0.385	0.310 ± 0.423	0.411	0.405
M02	0.595 ± 0.134	0.570 [0.426, 0.700]	0.513 ± 0.059	0.773 ± 0.288	0.253 ± 0.338	0.333	0.302

F2. D01 eye-tracking: model ROC-AUC vs the tracking-quality-only baseline (CIs: children resampled)

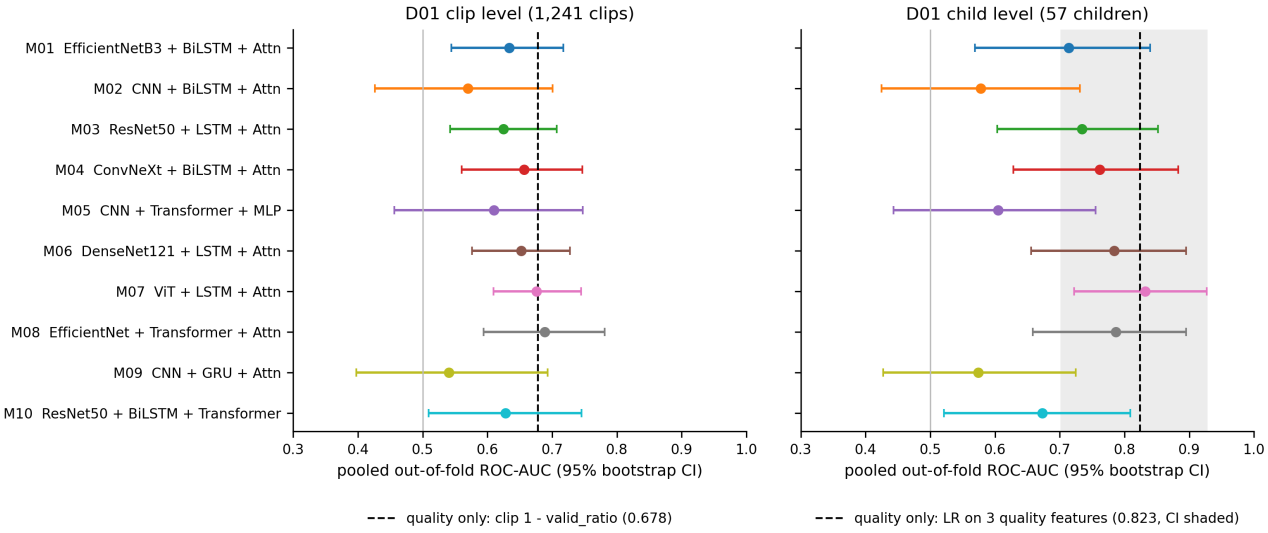


Figure 2. D01 pooled test AUC per model (95% CI) with the tracking-quality-only baseline as reference (dashed).

2.3 D02 facial images (image level)

Model	ROC-AUC mean ± SD	Pooled AUC [95% CI]	Bal. acc.	Sensitivity	Specificity	Brier	ECE
M04	0.931 ± 0.015	0.896 [0.884, 0.908]	0.845 ± 0.028	0.848 ± 0.074	0.841 ± 0.095	0.131	0.097
M07	0.924 ± 0.015	0.898 [0.887, 0.909]	0.832 ± 0.033	0.816 ± 0.085	0.848 ± 0.060	0.136	0.099
M08	0.903 ± 0.011	0.871 [0.857, 0.884]	0.819 ± 0.028	0.807 ± 0.036	0.831 ± 0.063	0.152	0.121
M01	0.901 ± 0.021	0.882 [0.870, 0.894]	0.812 ± 0.024	0.789 ± 0.060	0.834 ± 0.034	0.155	0.130
M06	0.901 ± 0.011	0.885 [0.873, 0.897]	0.811 ± 0.022	0.790 ± 0.047	0.832 ± 0.065	0.144	0.094
M03	0.890 ± 0.010	0.882 [0.870, 0.894]	0.804 ± 0.014	0.776 ± 0.027	0.832 ± 0.041	0.147	0.084
M10	0.885 ± 0.016	0.875 [0.862, 0.887]	0.789 ± 0.030	0.721 ± 0.096	0.856 ± 0.051	0.164	0.123
M05	0.881 ± 0.016	0.874 [0.862, 0.887]	0.795 ± 0.022	0.807 ± 0.081	0.783 ± 0.077	0.147	0.062
M09	0.875 ± 0.028	0.867 [0.855, 0.880]	0.778 ± 0.042	0.776 ± 0.073	0.780 ± 0.120	0.157	0.080
M02	0.867 ± 0.046	0.869 [0.855, 0.882]	0.785 ± 0.039	0.787 ± 0.052	0.784 ± 0.083	0.153	0.067

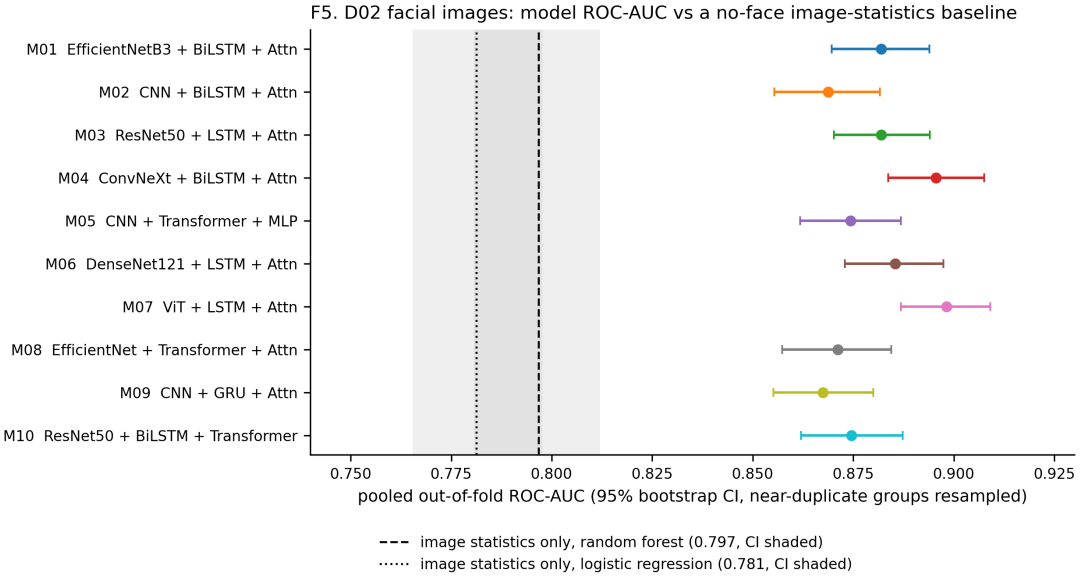


Figure 3. D02 pooled test AUC per model (95% CI) vs a baseline that uses only non-facial image statistics.

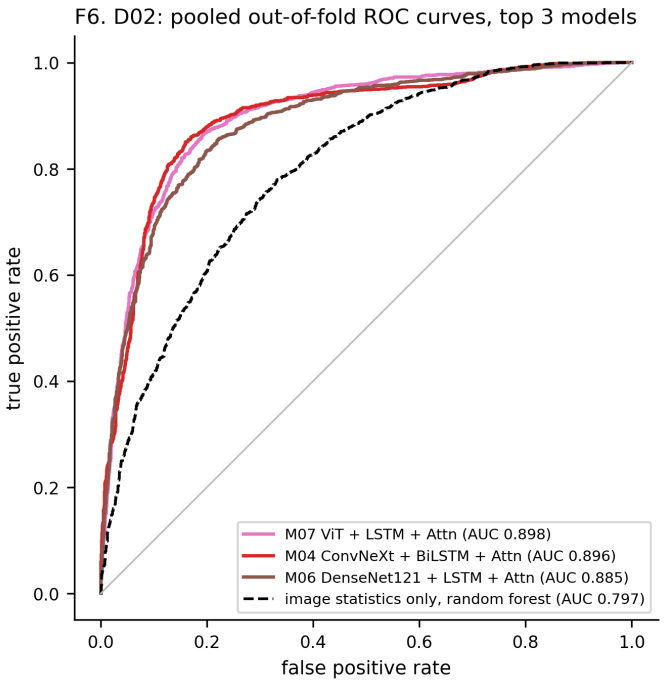


Figure 4. D02 pooled out-of-fold ROC curves of the three best models and the image-statistics baseline.

2.4 Tabular baselines (questionnaires)

Dataset / setting	Best model	ROC-AUC mean ± SD	Pooled AUC [95% CI]	Bal. acc.	Dummy AUC
D04 autism prediction	mlp	0.912 ± 0.031	0.905 [0.882, 0.926]	0.773 ± 0.067	0.500
D05 toddler, without A1-A10 / score	hist_gb	0.612 ± 0.074	0.610 [0.570, 0.652]	0.573 ± 0.073	0.500
D05 toddler, all features (label-definitional)	logreg	1.000 ± 0.000	1.000 [1.000, 1.000]	1.000 ± 0.000	0.500
D06 early autism	logreg	0.481 ± 0.065	0.482 [0.446, 0.519]	0.503 ± 0.057	0.500

D05 'all features' contains the A1-A10 items that define its label, so 1.000 is an upper bound by construction, not a result. D06 shows no signal (best pooled AUC below 0.5).

F10. Tabular baselines: ROC-AUC per model (dashed line = chance / dummy baseline 0.5)

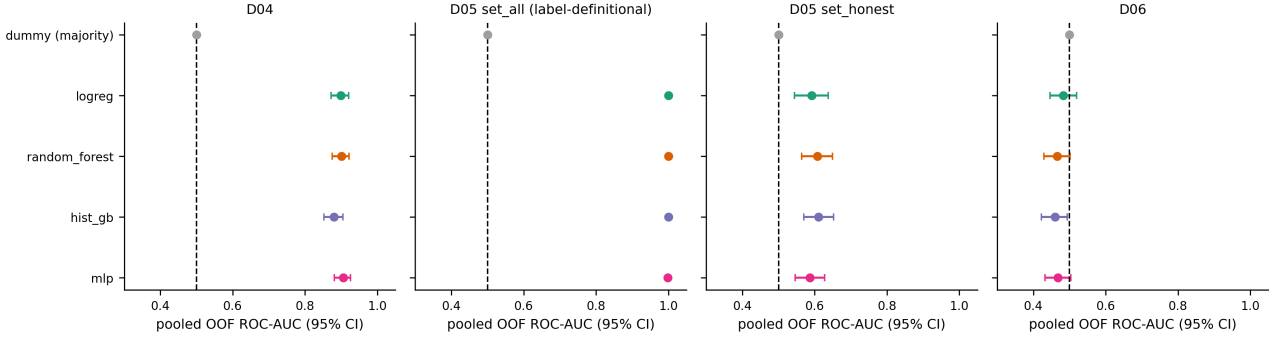


Figure 5. Tabular baselines: pooled AUC per model with 95% CI; dashed line = chance.

2.5 Confusion matrices of the best model per dataset (pooled out-of-fold, threshold 0.5)

Dataset	Best model	TN	FP	FN	TP
D01 clip	M08	572	232	195	242
D01 child	M07	27	3	10	17
D02	M07	1246	224	271	1199
D04	mlp	586	53	60	101
D05 set_all	logreg	326	0	0	728
D05 set_honest	hist_gb	172	154	278	450
D06	logreg	297	292	205	206

3. Results vs baselines: shortcut tests

A high test AUC is only meaningful if it cannot be explained by something simpler than autism-related patterns. Both datasets were tested against such shortcuts, without retraining any model.

3.1 D01: deep models vs tracking quality alone

A logistic regression on three per-child recording-quality features (valid-gaze ratio, blink ratio, tracking ratio), fitted per fold on training children only, reaches child-level pooled AUC **0.823** [0.701, 0.928]. After 1:1 matching of ASD and TD children on tracking quality (14 pairs), the quality score itself drops to 0.571.

Model	Child AUC [95% CI]	Model - quality baseline [95% CI]	DeLong p	Matched-subset AUC [95% CI]	Adds info (rule)
M01	0.714 [0.569, 0.840]	-0.110 [-0.265, +0.046]	0.161	0.556 [0.337, 0.765]	no
M02	0.578 [0.425, 0.731]	-0.246 [-0.417, -0.084]	0.003	0.469 [0.281, 0.663]	no
M03	0.735 [0.604, 0.852]	-0.089 [-0.233, +0.054]	0.230	0.587 [0.362, 0.796]	no
M04	0.762 [0.628, 0.883]	-0.062 [-0.196, +0.069]	0.351	0.592 [0.337, 0.816]	no
M05	0.605 [0.443, 0.756]	-0.219 [-0.398, -0.058]	0.008	0.515 [0.311, 0.709]	no
M06	0.784 [0.656, 0.895]	-0.040 [-0.151, +0.065]	0.478	0.597 [0.367, 0.781]	no
M07	0.832 [0.722, 0.927]	+0.009 [-0.098, +0.120]	0.876	0.724 [0.551, 0.883]	yes
M08	0.786 [0.658, 0.895]	-0.037 [-0.174, +0.107]	0.590	0.679 [0.469, 0.842]	no
M09	0.574 [0.427, 0.725]	-0.249 [-0.410, -0.090]	0.002	0.536 [0.316, 0.755]	no
M10	0.673 [0.521, 0.809]	-0.151 [-0.295, -0.007]	0.036	0.551 [0.347, 0.750]	no

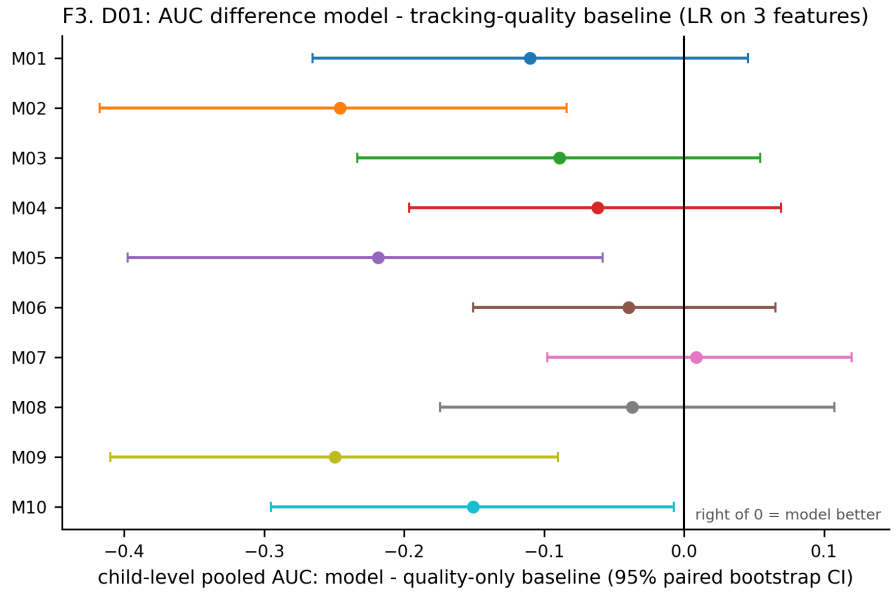


Figure 6. D01: child-level AUC difference model - quality baseline (95% paired bootstrap CI); right of 0 = model better.

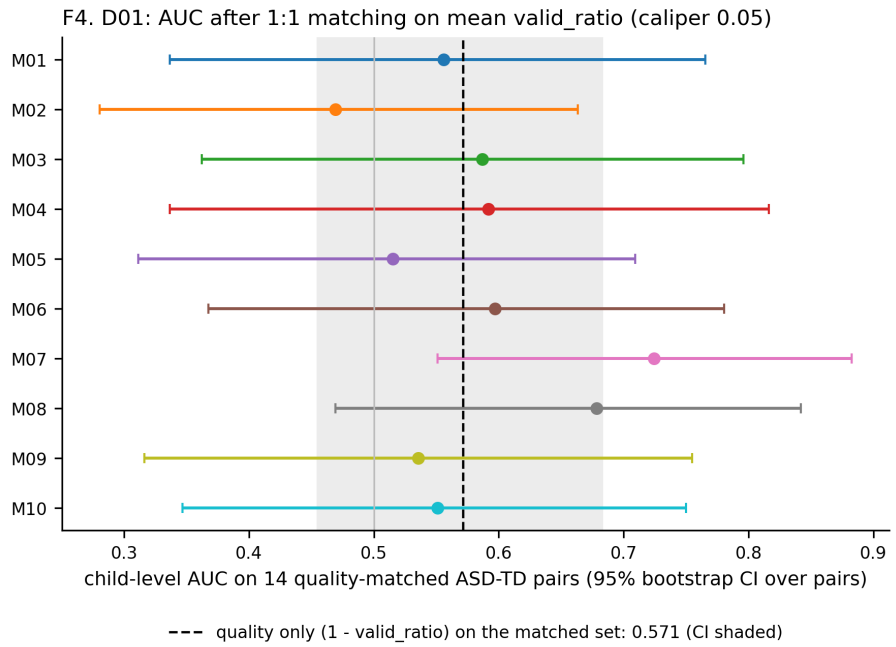


Figure 7. D01: AUC on quality-matched children (95% CI); grey band = quality-only score on the same children.

3.2 D02: deep models vs image artefacts and near-duplicates

A random forest on 17 non-facial image statistics (size, file size, JPEG quality, colour and brightness, border colour, sharpness) reaches fold-mean AUC **0.796** ± 0.019, so dataset artefacts alone separate the classes. Removing the 65 test images that have a near-duplicate (pHash <= 8) in the training images changes model AUCs by at most 0.002.

Model	Pooled AUC [95% CI]	Fold-mean AUC	Near-duplicates removed	Original 'test' folder - rest [95% CI]
M01	0.882 [0.870, 0.894]	0.901	0.899 (-0.002)	+0.020 [-0.022, +0.054]
M02	0.869 [0.855, 0.882]	0.867	0.865 (-0.002)	+0.040 [+0.001, +0.075]
M03	0.882 [0.870, 0.894]	0.890	0.889 (-0.002)	+0.028 [-0.009, +0.061]
M04	0.896 [0.884, 0.908]	0.931	0.929 (-0.002)	+0.050 [+0.018, +0.079]
M05	0.874 [0.862, 0.887]	0.881	0.879 (-0.002)	+0.027 [-0.012, +0.063]
M06	0.885 [0.873, 0.897]	0.901	0.899 (-0.002)	+0.003 [-0.040, +0.042]
M07	0.898 [0.887, 0.909]	0.924	0.923 (-0.001)	+0.028 [-0.007, +0.058]
M08	0.871 [0.857, 0.884]	0.903	0.901 (-0.002)	+0.041 [+0.003, +0.076]
M09	0.867 [0.855, 0.880]	0.875	0.874 (-0.002)	+0.041 [+0.002, +0.074]
M10	0.875 [0.862, 0.887]	0.885	0.884 (-0.002)	+0.040 [+0.003, +0.074]

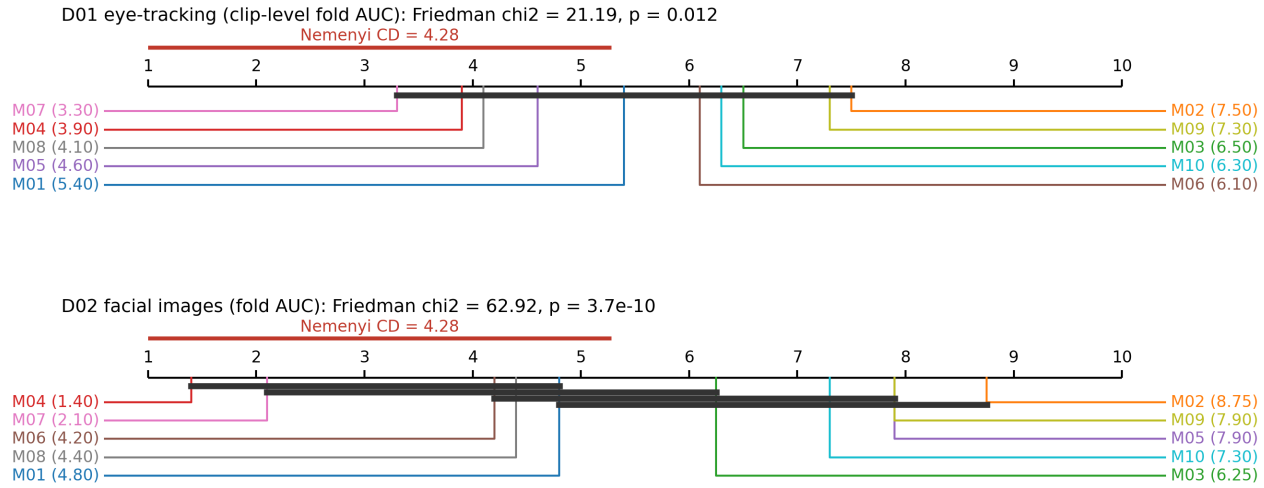
4. Statistical tests: are the models different?

Friedman test on the 10 folds x 10 models matrix of test AUCs, Nemenyi post-hoc test (alpha 0.05) and the Nadeau-Bengio corrected resampled t-test (best model vs each other model, Holm-adjusted).

Comparison	Friedman chi²	p	Nemenyi CD	Significant pairs (of 45)	Best average rank
D01 clip-level fold AUC	21.19	0.012	4.28	0	M07 (3.30)
D01 child-level fold AUC	7.69	0.566	4.28	0	M04 (4.00)
D02 fold AUC	62.92	3.7e-10	4.28	11	M04 (1.40)

On D02, M04 is significantly better than 7 of 9 models after Holm correction (M01, M03, M05, M06, M08, M09, M10); on D01 no pairwise comparison survives correction.

F7. Critical-difference diagrams



Average rank over 10 folds (1 = best). Black bars join models that are not significantly different (Nemenyi, alpha 0.05).

Figure 8. Critical-difference diagrams (average rank, 1 = best; bars join models that are not significantly different).

Pretrained vs from-scratch encoders (mean of model fold-mean AUCs, descriptive)

Analysis	Pretrained (7 models)	From scratch (3 custom CNNs)
D01 clip	0.658	0.619
D01 child	0.790	0.717
D02 image	0.905	0.874

F12. Reliability diagrams of the best model per dataset

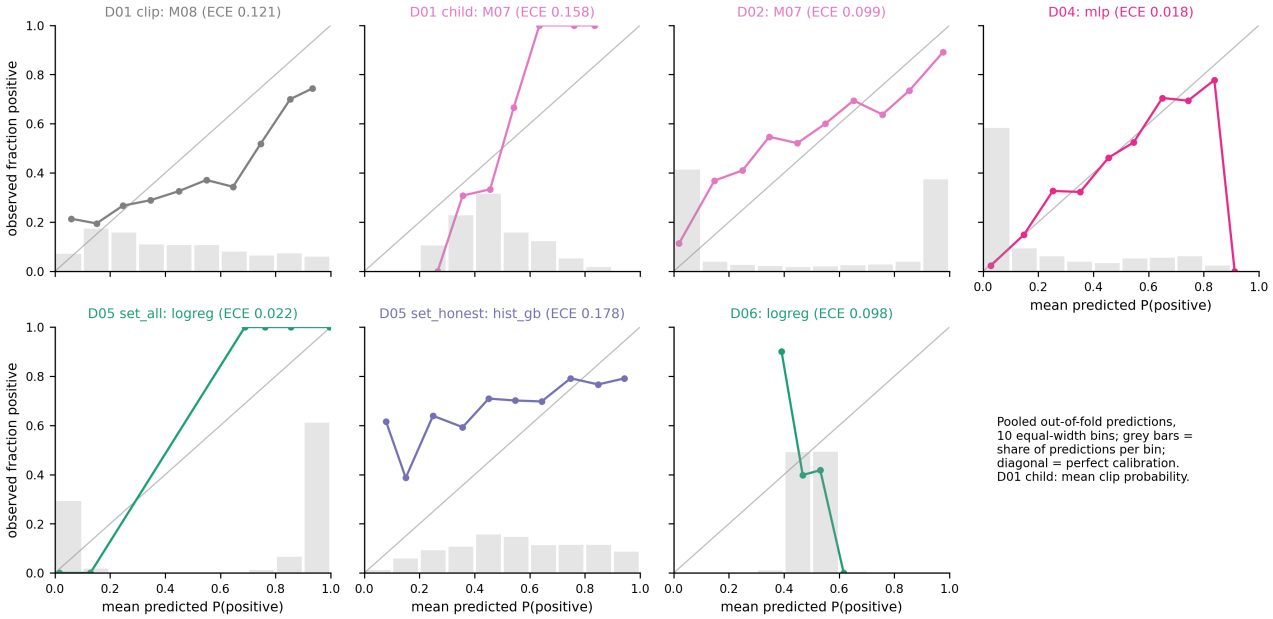


Figure 9. Calibration (reliability diagrams) of the best model per dataset.

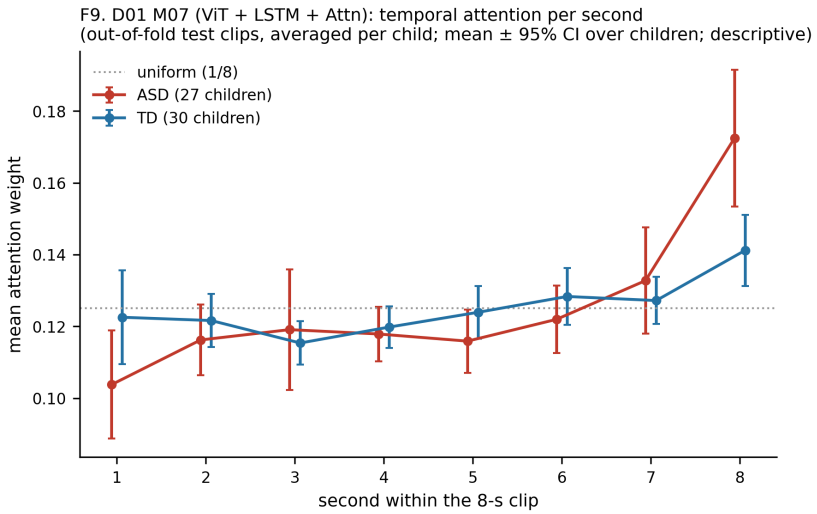


Figure 10. D01 M07 temporal attention per second, ASD vs TD (descriptive).

5. How the results themselves were tested

Before this document was produced, every stored result was re-derived by independent code (scikit-learn directly from the saved predictions, without the project's own metric code), checked for data leakage, and covered by the project's automated test suite.

Verification	Result
Automated test suite (pytest, run with results frozen read-only)	360 passed in 129.92s (0:02:09)
Per-run metrics recomputed from predictions (400 runs, 4,000 checks, tolerance 1e-6)	0 mismatches
Final master table recomputed (800 values: means, SDs, pooled AUC, ECE)	0 mismatches
Numbers quoted in the final report found in the result tables (1,977 numbers)	1,973 matched; 4 explained (derived or constants); 0 unexplained
Local snapshots of all 10 stages (code + checksums)	all intact, no file lost
Frozen result manifest and submission package	2,690 files hashed; zip 247 files, 5.61 MB

Leakage checks

Check	Result
Predictions of every run = that fold's test ids (400 runs)	passed
Every sample predicted exactly once per model	40/40 models
Train / validation / test disjoint by sample and by group (50 dataset-folds)	passed
No child (D01) or near-duplicate group (D02) in two test folds	passed
D02 near-identical image pairs (pHash <= 4) across folds	0 of 11
Excluded duplicate dataset D03 never used	passed
CARS score (label proxy) absent from every input	passed
D05 'honest' setting contains no A1-A10 items / score	passed
D04 unlabelled test.csv never used	passed

Each stage (S1-S10) ended with a written report, passing tests and a local snapshot before the next stage started.

6. Overall final result

In short: the hybrid models reach moderate test AUCs on eye-tracking (child level 0.574-0.832) and high test AUCs on facial images (0.867-0.931). But tracking quality alone reaches 0.823 on D01, and non-facial image statistics alone reach 0.796 on D02. The results therefore support **no claim** that the models learned autism-related gaze or facial patterns, and they are not clinical accuracy estimates.

What can and cannot be claimed (copied unchanged from the final report)

- Can be claimed: per-child tracking quality alone predicts the label about as well as or better than every deep model (quality_LR3 pooled child AUC 0.823; models 0.574-0.832); no model is significantly better than this baseline, and 4 models (M02, M05, M09, M10) are significantly worse (CI of the difference below 0). Model child scores fall as tracking quality rises, also within the ASD group.
- By the pre-specified rule, M07 adds information beyond tracking quality, via rule 2 only: matched-subset AUC 0.724 [0.551, 0.883] on 14 pairs. This is weak evidence: rule 1 fails (difference vs quality_LR3 +0.009 [-0.098, +0.120]); 10 models were tested without multiplicity adjustment (with no signal beyond quality, the chance that at least one of 10 independent one-sided 97.5% bounds lies above 0.5 is 1 - 0.975^10 = 22%; the models are correlated, so this is only a guide); the matched set is small and keeps some residual imbalance (quality-only AUC on it 0.571, SMD -0.16); the exploratory incremental analysis gives +0.012 pooled AUC.
- For M01, M02, M03, M04, M05, M06, M08, M09, M10: the D01 results are **not distinguishable from the tracking-quality shortcut**.
- Cannot be claimed: that the D01 models learned gaze behaviour characteristic of autism; that any D01 model outperforms a recording-quality measure; any clinical accuracy.
- Can be claimed: simple non-facial image statistics alone reach AUC 0.782-0.796, so the two classes differ systematically in image properties (colour, brightness, size/compression, sharpness); the deep models (0.867-0.931) score higher than this baseline, but how much of their performance comes from facial phenotype rather than such artefacts is not identified by these analyses.
- Near-duplicate leakage at pHash <= 8 does not explain the results (AUC change -0.002 to -0.001); different photos of the same child cannot be detected without subject IDs, so subject-level leakage is neither shown nor excluded.
- Models differ (Friedman p 3.7e-10); M04 has the highest mean fold AUC and is significantly better than 7 of 9 models after Holm (M01, M03, M05, M06, M08, M09, M10); pretrained backbones average higher than the custom CNNs.
- AUC is higher on the original 'test' images than on the rest for all 10 models (CI above 0 for 5): the original sub-folders are not exchangeable, a further sign of source heterogeneity.
- Cannot be claimed: that facial phenotype was learned; clinical validity; generalisation to clinical photos or other populations (web-collected images, no subject IDs, unknown diagnosis verification).
- D04: tabular models reach pooled AUC up to 0.905; D05 set_honest up to 0.610; D06 no model above 0.5 (max 0.482); D05 set_all = 1.000 is label-definitional.
- No result in this project is a diagnosis or a clinical accuracy estimate.

Why this motivates reliability-weighted multimodal fusion

Both modalities carry a signal whose source is uncertain: on D01 most of the separability is matched by recording quality, and on D02 by image properties unrelated to faces. A fused screening system should therefore not trust each modality equally. It should estimate, per child and per input, how reliable a modality is (for gaze: tracking quality, number of valid clips; for images: acquisition conditions, distance from the training distribution) and weight or reject that modality's prediction accordingly, and it should be evaluated against the same shortcut baselines used here. The present datasets do not contain the same children in both modalities (DD-11), so fusion itself could not be trained or tested in this project.

Limitations

- Sample size: D01 has 57 children (5-6 test children per fold); child-level CIs are wide and the matched analysis rests on 14 pairs.

- D02 has no subject identifiers: different photos of the same child can fall into training and test folds; only near-identical copies were grouped. Results are image-level, not child-level.
- D02 images were collected from the web with unknown diagnosis verification, age range, consent and acquisition conditions; image artefacts alone separate the classes.
- The custom CNNs were capped at 40 epochs by the fixed protocol and on D02 more than half of their runs reached the 40-epoch limit without meeting the early-stopping criterion (Stage 6 report), so their comparison with pretrained encoders is conservative for the CNNs.
- cuDNN benchmark mode was on: re-running a fold reproduces results only to about 1e-3 in the losses (Stage 4); the stored results are the reference.
- D01 clip-level and child-level results answer different questions; clips of one child are not independent.
- The tabular datasets are small public questionnaires; D05 set_all is label-definitional and D06 shows no signal.

Source files: reports/FINAL_RESULTS_REPORT.md (full report), results/final/*.csv (all numbers with source references), reports/STAGE_01..10_REPORT.md (stage records), logs/stage10_audit.txt (independent re-verification).

Appendix A. Per-fold test ROC-AUC

D01 clip

Model	F1	F2	F3	F4	F5	F6	F7	F8	F9	F10	Mean
M01	0.668	0.657	0.687	0.802	0.764	0.468	0.680	0.587	0.659	0.478	0.645
M02	0.666	0.737	0.767	0.599	0.689	0.415	0.637	0.507	0.574	0.358	0.595
M03	0.658	0.647	0.674	0.769	0.799	0.440	0.666	0.575	0.605	0.519	0.635
M04	0.598	0.664	0.683	0.795	0.813	0.631	0.711	0.679	0.704	0.548	0.683
M05	0.622	0.678	0.757	0.703	0.814	0.502	0.730	0.532	0.743	0.488	0.657
M06	0.557	0.621	0.791	0.785	0.766	0.508	0.661	0.573	0.609	0.522	0.639
M07	0.607	0.670	0.751	0.811	0.814	0.539	0.699	0.634	0.712	0.541	0.678
M08	0.732	0.604	0.784	0.907	0.794	0.651	0.680	0.605	0.669	0.438	0.686
M09	0.634	0.723	0.682	0.639	0.673	0.458	0.643	0.524	0.575	0.499	0.605
M10	0.603	0.650	0.649	0.879	0.837	0.447	0.663	0.582	0.556	0.516	0.638

D01 child

Model	F1	F2	F3	F4	F5	F6	F7	F8	F9	F10	Mean
M01	0.667	0.778	0.778	1.000	1.000	0.500	1.000	0.778	0.778	0.333	0.761
M02	0.667	0.667	1.000	0.778	0.833	0.167	1.000	0.444	0.889	0.333	0.678
M03	0.556	0.667	0.778	1.000	1.000	0.500	0.889	0.667	0.889	0.500	0.744
M04	0.556	0.778	0.778	1.000	1.000	0.833	0.889	0.889	1.000	0.833	0.856
M05	0.556	0.778	1.000	0.667	0.833	0.667	1.000	0.667	1.000	0.500	0.767
M06	0.444	0.667	1.000	1.000	1.000	0.667	0.889	0.778	0.889	0.500	0.783
M07	0.556	0.667	1.000	1.000	1.000	0.500	1.000	0.889	0.889	0.500	0.800
M08	1.000	0.667	1.000	1.000	1.000	0.833	0.889	0.778	0.778	0.333	0.828
M09	0.667	0.778	1.000	0.333	0.667	0.500	1.000	0.667	0.778	0.667	0.706
M10	0.444	0.778	0.889	1.000	1.000	0.500	0.889	0.778	0.667	0.667	0.761

D02

Model	F1	F2	F3	F4	F5	F6	F7	F8	F9	F10	Mean
M01	0.904	0.887	0.886	0.901	0.897	0.937	0.874	0.904	0.887	0.937	0.901
M02	0.883	0.885	0.869	0.899	0.743	0.873	0.868	0.872	0.866	0.908	0.867
M03	0.892	0.899	0.884	0.902	0.885	0.880	0.879	0.883	0.888	0.908	0.890
M04	0.928	0.930	0.909	0.914	0.950	0.950	0.916	0.940	0.930	0.945	0.931
M05	0.855	0.881	0.877	0.901	0.896	0.873	0.869	0.887	0.868	0.906	0.881
M06	0.887	0.902	0.885	0.904	0.903	0.907	0.903	0.893	0.903	0.924	0.901
M07	0.917	0.934	0.895	0.930	0.944	0.907	0.934	0.918	0.929	0.930	0.924
M08	0.891	0.903	0.902	0.915	0.892	0.926	0.898	0.909	0.893	0.900	0.903
M09	0.885	0.891	0.833	0.899	0.893	0.834	0.867	0.892	0.849	0.909	0.875
M10	0.858	0.898	0.869	0.889	0.891	0.877	0.890	0.871	0.901	0.909	0.885

Appendix B. Environment

Item	Value
python	3.12.3
torch	2.14.1+cu130
cuda	13.0
gpu_name	NVIDIA RTX A2000 12GB
driver	580.178.04
timm	1.0.30
tabular sklearn	1.9.1